

PROFILE

University of Queensland Computer Science graduate with Distinction from a top-50 global university. Python-first, backend-leaning software engineer building full-stack SaaS systems with FastAPI, Django, PostgreSQL, Docker, and Kubernetes. Applies deterministic validation, automated testing, persistent state, and observable failure handling to APIs and LLM-enabled features, with React as a supporting delivery tool.

CORE SKILLS

Backend: Python, FastAPI, Django, RESTful APIs, SQLAlchemy, SQLModel, Django ORM, Pydantic, Pytest
Data & Infrastructure: SQL, PostgreSQL, SQLite, Docker, Docker Compose, Kubernetes, GKE, GCP, Microsoft Azure, Linux, Git, Nginx
Frontend & AI: JavaScript, React, Vite, HTML5, CSS3, TailwindCSS, Google ADK, Gemini API, structured outputs, telemetry

PROFESSIONAL EXPERIENCE

Enfrasys Consulting | Cloud Infrastructure Intern

May 2022 - Aug 2022

Kuala Lumpur, Malaysia

- Configured BitLocker encryption and verified endpoint security baselines for **Bank Negara Malaysia** hardware, working within financial-sector data-protection requirements.
- Supported a phased enterprise Active Directory migration from on-premises infrastructure to cloud services, helping preserve service continuity throughout the transition.
- Provided Level 1 technical support for Microsoft 365 and Azure services, handling assigned requests within established SLA requirements.

SOFTWARE ENGINEERING PROJECTS

Interface Design Collection | Live Demos

2026

[HIVE & HUM Honey](#) | [Mugimori Kissa Coffee](#) | [Morrow Signal Token](#) | [Munch Club Cat Food](#)
Built four responsive concept sites with React or vanilla JavaScript, Vite, and custom CSS systems. Implemented ARIA tabs, dialogs, navigation, form feedback, live regions, reduced-motion paths, lazy media fallbacks, requestAnimationFrame pointer effects, IntersectionObserver reveals, and automated GitHub Pages builds.

AI Word Chain

2026

- Built a React/Vite client and FastAPI service around an ordered 3-node Google ADK workflow; used typed turn state and short-circuited creativity and move-generation nodes when deterministic validation failed.
- Kept format, chain, duplicate, dictionary, and scoring rules outside the LLM; validated Gemini candidates in code with bounded retries and local fallbacks, then persisted games, player statistics, and ordered traces through SQLModel and SQLite.
- Reconciled ambiguous network failures against authoritative `/game/{id}/state` instead of blindly retrying turns; exposed latency, token, error, and retry telemetry and verified 13 backend tests plus evaluation sets covering 22 validation cases, 22 creativity labels, 30 hard-mode sequences, and 10 adversarial inputs.

TableTap Scaling Project

2025 - 2026

- Engineered modular FastAPI routes and a PostgreSQL/SQLAlchemy model layer for Google OAuth, QR-linked menus, table-scoped ordering, status tracking, and owner dashboards; enforced valid order states, restaurant/table uniqueness, and cascading UUID relationships at the database layer.
- Containerized PostgreSQL, backend, and Nginx-served frontend services with Docker Compose; authored GKE manifests with persistent storage, resource limits, health probes, graceful preStop handling, managed ingress certificates, and HPA scaling from 3 to 10 replicas at defined CPU and memory thresholds.

GreenGuided Tourism Platform

2025

- Developed a Django/PostgreSQL itinerary engine that validates planner inputs, filters locations by travel-mode distance limits, applies persona and crowd-bias weights, orders stops with Haversine distance calculations, and builds pace-aware timelines.
- Implemented an ordered onboarding wizard that redirects out-of-sequence requests and requires transparency consent before the persona quiz; modeled locations, itineraries, visits, points, and researcher metrics, with fixture-backed tests for planner, badge, and analytics behavior.

AI-AUGMENTED SOFTWARE DELIVERY

- Uses AI coding agents as controlled engineering collaborators across SaaS and application projects: converts product intent into approved specifications, decomposes work into bounded tasks, delegates parallel investigation, and manages context against project evidence rather than unrestricted chat history.
- Customizes OMP operating rules and BMAD workflow extensions to define deterministic handoffs, context-free review, claim checks, and browser/API/runtime verification gates; configures repeatable workflows as reusable skills with explicit routing and validation contracts.

EDUCATION

The University of Queensland, Bachelor of Computer Science, Distinction (GPA 5.25 / 7.0), top-50 global university | Completed Jan 2026
HELP University, Diploma in Information Technology, GPA 3.71 / 4.0 | Jan 2021 - Dec 2022

CERTIFICATION & LANGUAGES

Microsoft Certified: Azure Fundamentals (AZ-900), 2022 | **Languages:** English (Professional Working), Chinese / Mandarin (Native), Malay (Conversational)